Marine Policy 109 (2019) 103702

Contents lists available at ScienceDirect

Marine Policy

journal homepage: http://www.elsevier.com/locate/marpol

Social equity and benefits as the nexus of a transformative Blue Economy: A sectoral review of implications

Andrés M. Cisneros-Montemayor [a,*], Marcia Moreno-Báez [b], Michelle Voyer [c], Edward H. Allison [d], William W.L. Cheung [a,e], Margot Hessing-Lewis [f,g], Muhammed A. Oyinlola [a,e], Gerald G. Singh [a], Wilf Swartz [a], Yoshitaka Ota [h]

[a] Nippon Foundation Nereus Program, Institute for the Oceans and Fisheries, University of British Columbia, Vancouver, Canada
[b] Department of Environmental Studies, University of New England, Biddeford, ME, United States
[c] Australian National Centre for Ocean Resources and Security (ANCORS), University of Wollongong, Australia
[d] School of Marine and Environmental Affairs, University of Washington, Seattle, USA
[e] Changing Oceans Research Unit, Institute for the Oceans and Fisheries, University of British Columbia, Vancouver, Canada
[f] Hakai Institute, Quadra Island, Canada
[g] Institute for the Oceans and Fisheries, University of British Columbia, Vancouver, Canada
[h] Nippon Foundation Nereus Program, School of Marine and Environmental Affairs, University of Washington, Seattle, USA

A B S T R A C T

The term 'Blue Economy' is increasingly used in various marine sectors and development frameworks. For it to be a truly useful approach, however, we argue that social benefits and equity must be explicitly prioritized alongside environmental and economic concerns. This integration of social dimensions within the Blue Economy is required to ensure that marine economic sectors contribute to achieving sustainable development goals. We review what an equity-focused 'Blue Economy' might mean for some established and emergent marine sectors and note existing guidelines that may be used for incorporating these aspects into planning. Moving towards a Blue Economy does not only imply developing emerging sectors in undeveloped areas; larger challenges will be found in transforming industries that already have significant economic and livelihood contributions despite concurrent social and environmental concerns. A 'marine industrial revolution'—as the Blue Economy has sometimes been understood—cannot achieve sustainable development and well-being if it does not avoid the widespread negative social and ecological impacts of historical development pathways. A concerted effort is therefore necessary to design and implement inclusive and equitable policies as an integral part of a Blue Economy that is transformative and not only expansive.

1. Introduction

The Blue Economy has been proposed as an ocean-based counterpart to the Green Economy development plan [1] to improve human well-being and social equity while achieving environmentally sustainable economic growth [2]. Like the more recent and broader UN Sustainable Development Goals (SDGs) [3], this goal is aspirational but articulates a clear vision of a world most would like to see. This (ideally) distinguishes "Blue Economy" from the "Ocean (or Marine) Economy," which comprises any and all economic activity (both public and private, but excluding non-market value) related to oceans [4], and "Blue Growth," the expansion of this Ocean Economy in a market economy sense with some implied observance of environmental sustainability [5]. This paper argues that a focus on social equity is what can make 'Blue Economy' a new and more meaningful approach. This is not merely an instrumental argument, i.e., 'social needs must be addressed to achieve sustainability or economic growth.' Social well-being is also goal in and of itself, as stated throughout the SDGs [3]. This is discussed using examples of 'Blue Economy' sectors, noting some existing guidelines that could be useful for future planning that integrates social considerations in ocean policy.

Over time, 'Blue Economy,' 'Ocean Economy,' 'Blue Growth,' and other similar terms are increasingly being used interchangeably [1,5]. 'Blue Economy' discourses have thus tended more towards business-as-usual planning, focusing on technological and financial aspects of economic growth rather than the original and more holistic social-ecological perspective that focused on human well-being or equity specifically [1,6,7]. As is further discussed below, this interpretation undermines 'Blue Economy' as a distinct (and thus useful) new term and approach. Research over the past decade has found an initial divergence and current cementing of different understandings of 'Blue Economy' as discussed in international forums, with oceans being

* Corresponding author.
E-mail address: a.cisneros@oceans.ubc.ca (A.M. Cisneros-Montemayor).

https://doi.org/10.1016/j.marpol.2019.103702
Received 8 July 2019; Received in revised form 4 September 2019; Accepted 23 September 2019
Available online 3 October 2019
0308-597X/© 2019 Elsevier Ltd. All rights reserved.

viewed as 1) natural capital (led by international conservation organizations); 2) integral to coastal livelihoods (led by development organizations and mainly focused on recognizing the contribution of artisanal fisheries to the global economy); 3) good business (led by intergovernmental agencies and business associations); and 4) drivers of innovation (led by the investment and finance sector) [6]. Crucially, all of these discourses are situated at different points within overall market-based economic assumptions and goals, with social sustainability and well-being, if mentioned at all, implicitly assumed to follow from economic growth in ocean industries. Given that the Blue Economy has been advanced as comprising multiple sectors, a breach of social trust in one of these sectors can undermine trust in the entire term and approach [8]. This is evident in wary or hostile attitudes towards the term from artisanal fishers fearing dispossession of resources as a result of 'Blue Economy' frameworks (e.g. Ref. [9]) that could be interpreted as mere 'ocean-grabbing' [10].

This continued shift away from social equity and human well-being as a proximate goal is evident in planning proposals for ocean development. For example, marine fisheries—the most important ocean industry for jobs and food provision [11], particularly through the artisanal sector [12]—are sometimes excluded from ocean development plans (e.g. Ref. [13]) due to their low perceived potential for investment returns or overlaps with existing regulations [5,7]. Specific social considerations, such as equitable access to resources, sharing of benefits, and adherence to human rights and labor laws, are similarly absent from recent high-profile documents on the 'Blue Economy' (e.g. Ref. [14]). Most seafood certification programs, which have gained recognition and increasingly engage in developing regions, remain focused on environmental sustainability, rather than social sustainability, equity or fairness [15]. This study argues for and contextualizes the importance of social equity and benefits within the Blue Economy, highlighting how an integration of these concerns changes traditional planning but is essential to distinguish the approach from business-as-usual economic growth or environment-focused initiatives.

Here, we highlight potential benefits, synergies, and limitations in the establishment of a Blue Economy defined as ocean resource-based development that is socially equitable, environmentally sustainable, and economically viable. This view comprises various conceptualizations of 'Blue Economy' developed by Silver et al. [1] and adapted by Voyer et al. [6] (i.e., oceans as natural capital; livelihoods; good business; drivers of innovation). The critical factor in the definition used here, however, is that a fundamental change—namely, the incorporation of social equity concerns—in resource use and development is required to ensure human security amidst ongoing global challenges. This necessarily entails addressing pressures that occur globally but impact specific communities and regions very differently. Examples include shifting distributions of marine biodiversity and sea level rise as a result of climate change [16], and inequitable economic development within increasingly globalized markets and supply chains [17]. Importantly, recognizing and addressing these uneven impacts is central to the UN Sustainable Development Goals, i.e., "no one will be left behind" [3]. Nevertheless, most current documents relating to the Blue Economy have focused on the economic aspects of marine-related industries [4], but not necessarily environmental or social ones.

Social equity here refers to: the recognition and fair treatment of all groups that would benefit from or be impacted by existing or nascent ocean industries; their inclusion in development plans and policies that would affect them, and; the achievement of a more just distribution of benefits and burdens from these industries [18]. In practice, it is likely that the main focus for planning would be on procedural justice (a fair process of development and establishment for all those involved in or affected by marine industries) rather than distributional justice (a fair distribution of costs and benefits). Nevertheless, the latter may be expected to be met within a truly inclusive design and implementation of development policies, where historically and currently marginalized groups have a say in how "fairness" is defined [19]. Note, however, that

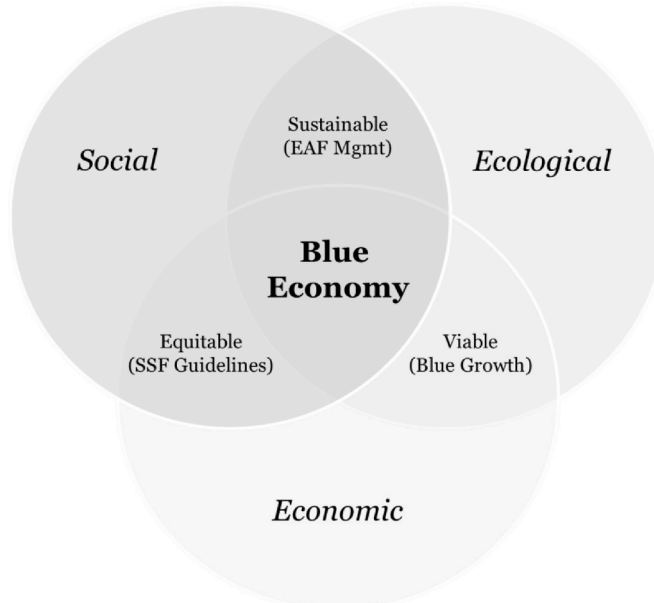

Fig. 1. Relationship of a Blue Economy (as defined here) with other governance goals and considerations. Parentheses indicate examples of frameworks and guidelines that could be integrated in a Blue Economy: Ecosystem Approach to Fisheries Management [26], FAO Small Scale Fisheries Guidelines [25], Blue Growth strategies [14].

measuring the distribution of benefits or costs across stakeholders is relatively straightforward but it is much more difficult to evaluate procedural equity and resulting social, environmental, and economic outcomes, however these may be defined [20]. This is particularly true when historical inequities between stakeholders of ocean resources need to be addressed.

These methodological and conceptual challenges may partially contribute to the lack of information on social themes and indicators despite considerable interest in the market contributions (e.g., revenue and employment) of ocean sectors. The current yearly contribution of ocean economic sectors to global GDP has been conservatively estimated at US$1.5 trillion, supporting some 31 million jobs, and is projected to double by 2030 [11]. This estimate notably excludes artisanal marine fisheries, a sub-sector thought to employ around 22 million people, and the formal and informal seafood processing sector employing an additional 210 million people [12]. Estimates of economic value have been carried out at the national level to aid in policy planning, including in Bangladesh, China, and the USA [14,21], and potential benefits from specific sectors have furthermore been projected based on expected technological advancements [11] or improvements to current governance [22,23]. Much less attention (aside from mainly academic case studies) has been placed on the equitable distribution of these benefits, and still less on aspects of social equity surrounding the development and implementation itself of marine sectors. This may stem from disciplinary boundaries as noted above, or from concerns that prioritizing social considerations would detract attention from ecological conservation objectives [20]. And yet, there is much evidence that these social aspects, from the observance of human rights to governance capacity to community cohesion (among many others), are key to achieving any management objective, including ecological ones [24].

Incorporating social equity into the Blue Economy approach makes it a more useful framework for achieving sustainable development goals, and this can be aided by existing guidelines for multi-objective development (Fig. 1). For example, this could include elements from the FAO Small Scale Fisheries Guidelines [25] to address equitable economic development, the Ecosystem Approach to Fisheries Management [26] to achieve sustainability, and the Blue Growth [14] framework for

ensuring that economic growth is environmentally sustainable, to name only a few. Given our definition of a Blue Economy as conceptualized in Fig. 1, overlaps between any two of the three areas are certainly beneficial (e.g., 'Sustainable,' 'Equitable,' 'Viable'), but only the inclusion of all three would be considered a Blue Economy. A focus on social equity may help bridge the common divides between economic and environmental goals in development planning, yet social equity and human well-being should not be viewed as means to an end but as the overarching goal.

This integrative approach to a Blue Economy fits well with the SDGs, that also explicitly call for synergistic and collaborative policies [3]. This Blue Economy approach also does not assume or require that multiple sectors overlap in an area (though this may certainly be desirable), but that any ocean sector is developed and operates in accordance with best practices to promote social equity and environmental sustainability as well as economic viability (Fig. 1). In this sense, while marine spatial planning tools can be very helpful, achieving a Blue Economy would require important inputs from social and political sciences and research.

1.1. Transitioning to a Blue Economy: a brief sectoral review

Technological advances are rapidly facilitating ocean-based economic growth, but marine industries are neither new nor limited in scale. Despite the lack of implemented Blue Economy (*per se*) projects to date, most sectors that would fall within the rubric of a Blue Economy, including ocean energy, blue carbon, marine bioprospecting, mariculture, marine tourism and fisheries, already occur throughout the world [27–32]. This is important because moving towards a Blue Economy not only implies developing innovative and emerging sectors and practices in undeveloped areas; larger challenges will be found in retrofitting and transforming industries that already have significant economic and livelihood contributions, however bad their social and environmental performance may currently be.

Following from the discussion above, we focus here on a subset of sectors that would likely be included within a Blue Economy as defined here: bioprospecting, blue carbon, blue energy, ecotourism, fisheries, and mariculture. We include these sectors as examples and do not intend to address here which sectors should or should not be included in a Blue Economy plan (for extensive reviews of potential sectors, see Refs. [4, 33]). However, we do argue that although all ocean-related industries, from renewable resource use to technology development to shipping and offshore oil drilling, etc., fit within broader Ocean Economy accounting and planning [4], a Blue Economy should be viewed as a particular form of development that very explicitly incorporates long-term social as well as environmental and economic goals and relies on the functioning of ocean and coastal ecosystems (i.e., Fig. 1). In that sense, industries that are by definition unsustainable—such as extraction of non-renewable resources—can and should reduce their negative environmental and social impacts, but including them within a 'Blue Economy' as defined here undermines the framework as a novel development approach. Some sectors do not necessarily require healthy ocean ecosystems but are very clearly linked to wider efforts for sustainability and climate change mitigation; we therefore include one such example, Blue Energy, in our reviews.

The main focus of this section is to highlight how a Blue Economy with social equity at its core would look like for some examples of ocean sectors. To a lesser extent, because it is already well-established in planning [34], we also consider some key aspects of environmental sustainably related to these sectors. We additionally recognize that economic viability is crucial for any industrial development, and the potential for different ocean sectors can sometimes be overstated in ocean development projections, particularly at the regional level (i.e., not all regions have ample potential for every marine industrial sector).

With this in mind, the following brief sectoral reviews provide 1) an overview of the current state of the sector and its key resource requirements, and 2) social equity (and, more briefly, ecological) concerns that would need to be addressed under our definition of a Blue Economy. Throughout this discussion we note some existing guidelines that could help establish or transition into socially equitable ocean sectors.

1.1.1. Bioprospecting

Bioactive substances in marine organisms have been used by humans for thousands of years and are a well-established source of modern compounds used in cancer and pain treatments, anti-fouling paints, and anti-bacterial drugs, among many other pharmaceutical and industrial applications [27]. There has been renewed interest in bioprospecting with a shift in focus from bioactive compounds to genetic information [35], particularly given modern capacity to synthesize material of interest in a laboratory, without needing further harvest of wild organisms [27]. Similarly, mariculture and bioengineering can increase supply of species of interest without adding pressure on wild stocks [36,37]; of course, strict safeguards are required to prevent releases of non-native organisms into the environment, particularly if these have been genetically modified.

The above, however, implies that private companies engaging in bioprospecting are able to patent and produce their own compounds without needing to share further benefits with the nation or regions where the organisms were found and that rarely have the capacity to engage in the industry themselves. Indeed, a single private firm owns 47% of global gene patents from marine organisms, and 98% of all such patents are owned by firms in 10 countries [38]. There is evidence to suggest that the drafting of the Nagoya Protocol, intended to increase benefit-sharing of marine genetic resources, may have led to a race to patent sequences before it could be implemented [38]. An additional economic constraint is that tens of thousands of compounds need to be screened to find one promising lead, and only 1 in 50 of these may result in a marketable product after some 15 years [27]. The resulting concentration of benefits from marine genetic diversity then is partly due to the fact that only a few large corporations have the capacity and resources to invest in this long-term research and absorb costs and risks along the way (a similar dynamic of concentration has been shown to occur in fisheries with tradeable quotas [39]).

A risk here of commodifying biodiversity is that if conservation interest hinges on the potential for continued economic benefits from bioprospecting, problems arise when these benefits can be produced in a laboratory. This makes initial permission and benefit-sharing agreements between private firms and coastal states vitally important, particularly given the language of the Convention on Biological Diversity [40] that affirms the right of states over their resources *in situ* but does not force any specific forms of benefit sharing from resulting products. This creates incentives for circumventing origin states, not necessarily by collecting without authorization but by taking advantage of traditional and Indigenous knowledge (often with no further benefits to original knowledge holders [41]), seeking other sources for material, or creating synthetic compounds. One alternative may be to decrease focus on direct financial benefits and rather seek partnerships and collaborations to increase local capacity, for example by engaging in technology transfer, student and professional exchanges, and other research collaborations [35,41]. In the context of avoiding social inequities, these types of arrangements can benefit nations much more widely and can also decrease potential issues of corruption or misallocation of direct cash transfers.

Given the complex legal, jurisdictional, and technological aspects of marine bioprospecting, a strong focus on transparency and support for the inclusion of least-developed countries in international regulatory frameworks is crucial for achieving a more equitable distribution of benefits among private firms, coastal states, and the public at large [27, 38,42]. These benefits include revenues from pharmaceutical and industrial products, but also much broader technological knowledge and capacity, and the products themselves (e.g., medicines).

1.1.2. Blue Carbon

'Blue Carbon' [43] emphasizes the historically underappreciated value of coastal vegetated habitats—seagrass meadows, mangrove forests, and tidal salt marshes (SMM)—as efficient carbon sinks with global significance for climate regulation [44–48]. Global syntheses have revealed that these systems can equal or outperform terrestrial ecosystems in terms of their relative sedimentary carbon burial rates [49]. New research also suggests that coastal shelf sea sediments are major stores of carbon (just as terrestrial soils are) that may rival terrestrial forests in total carbon storage capacity [50,51]. These carbon stores may, however, be somewhat less amenable to management, except by the regulation of fishing activities that disturb sediments, such as bottom-trawling [51]. These emerging findings have prompted international efforts to integrate Blue Carbon within climate mitigation policy [52] and economics-based conservation and restoration strategies [53].

The concept of Blue Carbon, i.e., conferring payments for the maintenance of SMM habitats equivalent to their carbon storage rates has built on existing climate policy and terrestrial carbon management. It is thus able to incorporate lessons from these initiatives, especially pertaining to critical discussions of procedural equity. To translate existing policies to the oceans, broad-based international policies (e.g., UN Framework Convention on Climate Change, UNFCCC) are being leveraged to support Blue Carbon initiatives [54] and other international and national policies can be similarly adapted [55,56]; for example, REDD + policies and programs can include mangrove habitats under the existing definition of 'forests' to facilitate Blue Carbon pilot projects [57]. A growing number of countries (~28), now list Blue Carbon ecosystems in their nationally determined contributions (NDCs) for the Paris Agreement [58], and integrate Blue Carbon into marine planning [59], though this has not included habitats beyond SMM [60].

Blue Carbon initiatives are wide-ranging in their strategies for habitat conservation but have been criticized for limited consideration of social equity concerns at local, national, and international scales. These include lack of access to funds across countries, limited participation of local community members in project planning, implementation, and benefit sharing, and the overarching human justice dimensions of projects that may push out traditional users (e.g. Ref. [61]). In response, equity considerations have been at the forefront of REDD+, including the implementation of safeguards and reporting to counter potential negative consequences [62]. The Blue Carbon sector continues to build on this, and many case studies from developing countries have incorporated equity considerations into practice, including prioritizing local livelihood benefits, and community involvement in planning and implementation [63]. Furthermore, the Blue Carbon scientific community has responded to the technical research challenges of this work by adapting methods to encourage community groups to participate in carbon inventories and the requisite monitoring components of many Blue Carbon projects [64]. Social justice and equity dimensions have also been included via compliance mandates within carbon markets (e.g., the Kyoto Protocol) [57]. This includes the regulated compliance-based market, but Blue Carbon projects have been primarily associated with the expanding, deregulated voluntary market [65]. For instance, the verified carbon standard (VCS), which accounts for more than half of the voluntary carbon market, considers Climate, Community and Biodiversity Standards (CCBS), and 'Social Carbon' co-benefits [66]. Emissions accounting and verification are strictly enforced, but though 'Gold Standards' have been developed to strengthen socio-economic outcomes they are not a necessity across the voluntary carbon marketplace, and would benefit from further inclusion within the compliance market [67].

Carbon markets, as a tool for stimulating the Blue Carbon sector, should be evaluated critically, as they may also present limitations for this sector within the larger Blue Economy. First, Blue Carbon initiatives currently focus on habitats that demonstrate high potential for carbon storage [60], limiting participation in Blue Carbon projects for regions without these areas or with lower natural capacity. Emerging data from temperate latitudes (e.g. Refs. [68–70]) suggests that carbon sequestration may be lower than global averages, and spatially variable. This may limit the potential profitability of projects given local constraints and research costs, and emphasizes the need for local carbon stock inventories and more holistic policies and financing mechanisms, for example to incorporate the co-benefits of SMM conservation for fisheries, ecotourism, mariculture, and non-quantifiable human well-being. A broadening of Blue Carbon definitions, however, may not fit well within regulated carbon markets, where strict emissions accounting guidelines are needed to ensure compliance and avoid leakage (i.e., "exporting" of impacts that results in no net gain). Payments for Ecosystem Services (PES) schemes have been criticized for overly commodifying nature, ignoring the complexities of social-ecological systems, and not incorporating non-market values, yet these frameworks can also provide operational tools for evaluation of environmental and social outcomes [18]. In the case of Blue Carbon, understanding and evaluating all market and non-market components is key to weighing potential costs and benefits of local projects. This is especially important for groups that have historically been excluded from governance and access to marine resources, and international economic and policy support could facilitate their direct engagement and decision-making [54].

1.1.3. Blue Energy

The single most important long-term strategy to mitigate climate change and its negative impacts is to reduce fossil fuel emissions [16], and thus one of the Blue Economy sectors of highest interest and projected profit growth is renewable ocean energy, or 'blue energy' production including offshore wind and tidal energy [11,32]. Approximately 40% of the world's population lives near a coast [71] and, particularly for urban areas, ocean turbines can be placed close to demand where there would be little space onshore [32]. Blue energy can furthermore be highly useful for developing regions of the world, including in otherwise developed countries, as "last mile" electrification using conventional grids can be difficult or costly [72]. Solar power has been effective in serving such areas, and blue energy sources could be a complement in places with seasonal shortages of solar energy, or limited area on land, e.g., in Small Island Developing States. This could also be further focused on providing energy for specific local needs like water purification to serve coastal communities in arid climates [73].

Ocean energy, relying primarily on tidal currents, has not yet expanded commercially in the same way as offshore wind projects, but the core technology is similarly proven (i.e., water turbines in dams) and it has a marked advantage over other forms of renewable energy because tidal flows are highly consistent over time [74]. In addition to wind or current speed and consistency, current limitations to ocean renewable energy have to do with the technical challenges and costs of installation and of maintenance turbines themselves [74]. Existing projects are therefore sited in shallow nearshore areas, though there are ongoing efforts to improve cost-efficiency and capacity, including floating wind turbines that would allow for deployment in deeper waters and automated monitoring and maintenance technology [75].

Aside from economic and technical considerations, however, a key challenge for implementing blue energy projects has been incorporating the preferences of local residents who may not wish to live next to expansive energy production sites, particularly when placed in areas that are relatively undeveloped [76,77]. Other users of ocean space can also be affected, including fishers, recreational boaters, shipping and mariculture operations. This, of course, reflects the realities of spatial planning, particularly involving relatively new industries, that must consider other industrial sectors, the various risks associated with new industrial development, and public perceptions of impacts, which are legitimate whether they be fishers losing fishing grounds or residents concerned about their ocean view [77]. An added challenge is that there are often stark differences in acceptability for renewable ocean energy

projects across wider public and government levels. In these cases, preemptive inclusion and collaboration in planning, and a transparent decision-making process can help avoid costly delays in development due to valid challenges by affected stakeholders. Particular care must be taken to ensure that local stakeholders receive some benefits, and recent examples highlight creative ways that include cheaper energy for local residents but also the creation of funds for targeted investments (e.g., installation of high-speed internet as part of project construction) using profits from developments [77].

1.1.4. Ecotourism

Marine ecotourism, formally defined, requires interactions with ocean life (unlike, say, surfing or sailing) and is generally assumed to incorporate environmental education and conservation, and local economic benefits [78]; it therefore fits well with our definition of Blue Economy. Marine ecotourism is by now a well-established global industry, generating well over US$50 billion in expenditures and over 1 million jobs around the world [28]. The most prominent marine ecotourism industries are recreational fishing, scuba diving and snorkeling, and viewing of sharks and marine mammals [28,79,80], though there are numerous other initiatives such as coasteering, coastal hikes and kayak trips specifically advertising marine ecosystems. Marine tourism has been identified as a key driver of global ocean economic activity, though there is little specific mention of the differences—in terms of scale or social and environmental benefits—of ecotourism and mass coastal tourism [11]. This is problematic given that the latter is often unsustainable and not specifically focused on social goals; indeed, there are cases where it has led to serious social harm and inequalities. These can include the appropriation of land and coastlines by tourism developments, the marginalization of local communities and appropriation of cultures, and potential regional dependency on tourism that hinders development of other sectors and communities. All these outcomes can contribute to inequalities, environmental degradation, and conflicts between tourists, locals, and incoming developers and workers [81]. Acknowledging the many variations of marine tourism industries and outcomes around the world, we focus here on ecotourism as a form of tourism that is in principle more appropriate under a Blue Economy.

Available evidence suggests that, aside from the governance capacity to ensure social and environmental benefits are sustained and equitably shared, there are two fundamental factors in the successful establishment of ecotourism operations. One is an existing tourism industry and/or infrastructure that supports tourist arrivals to the region. Although ecotourism continues to grow throughout the world [82], there are few places where ecotourism ventures, which almost by definition are relatively small, can by themselves be the primary driver of infrastructure investments large enough to attract and support large numbers of tourists. Rather, ecotourism can engage existing tourists during their trips or form (sometimes a very important) part of their set of planned activities. A second vital factor in attracting ecotourists to a particular site is either a significant and predictable occurrence of particular species (e.g., orcas, grey whales, whale sharks, billfish, sealions, coral reefs) or a large array of different species (e.g., reef-associated fishes, shorebirds, marine mammals). There are of course many other contextual factors that contribute to successful ecotourism industries, but the two highlighted here have been found to be useful for forecasting their general economic potential [83].

Ecotourism can provide significant social, economic and ecological benefits, but must be carried out in specific manners to achieve them [84,85]. For example, overcrowding in popular nature tourism destinations has been shown to be detrimental to reefs [86], including when novice divers or snorkelers accidentally break corals with their hands or equipment, or when they touch or take marine life (e.g., soft corals, shellfish). Likewise, aggressive pursuit of marine mammals and sharks by tour operators can potentially harm individual animals and populations [87,88] and lead to incorrect perceptions of wild animals among tourists [89].

Guidelines for best practices have been developed and are being implemented in sites around the world [90]; these can of course be modified according to local contexts, but fundamental norms to ensure the safety of both animals and tourists should not be relaxed. When it comes to observing marine life in the wild, most guidelines focus on not touching wildlife, allowing moving animals to maintain their course, limiting vessel time around animals, and limiting the number of people or vessels in a site at a time. Instituting environmental education as a core part of the ecotourism experience has been shown to increase awareness and support for conservation among participants, regardless of their prior knowledge or perceptions [89]. Similarly, involvement of locals in running ecotourism operations promotes a more equitable distribution of economic benefits and increases the entire community's awareness of environmental conservation and potential alternative livelihoods [91], and may aid in diversifying tourist segments to avoid overdependence on tourists from specific regions [81].

1.1.5. Fisheries

Capture fisheries support the most marine-related jobs throughout the world (some 260 million [12]); and have historically provided most seafood for human consumption [92], including through formal and informal local and international markets [93]. Fishing industries operate in every ecosystem type, including those focused on single high-value (or volume) species or diverse species throughout seasons; fisheries thus support employment, food security and sovereignty, and cultural identities in every maritime nation [94,95].

Despite its contribution to communities and economies globally, key challenges and a general perception that fisheries are difficult or impossible to properly manage at a global scale have perhaps contributed to the exclusion of capture fisheries as key future growth sectors in some 'Blue Economy' development plans, for example in the EU Blue Growth Strategy [96]. The negative effects of poorly managed fisheries are extensive and well documented, and include declines in marine species and ecosystems [97,98], social and economic negative impacts through questionable subsidy programs [22,99,100], and conflicts and human abuses [101,102]. Further, projections imply modest gains even in the best of scenarios, for example, a global increase in US$70 billion per year in landed value if every single fishery operated optimally [22]. Even within Blue Economy plans, a focus on high-tech and increased industrial development carries risks that artisanal fisheries, which support some 90% of all fishing-related jobs [12], will be excluded or otherwise marginalized from planning and large-scale investments. There is clear precedent for this, with industrial fishing firms receiving 85% of global fisheries subsidies despite providing only 10% of jobs and ~30% of fish for human consumption [103].

Fisheries are nonetheless vital to include within Blue Economy planning, for three main reasons. First, people occupied in fisheries directly and indirectly often represent the bulk of coastal populations, particularly in remote regions in developing countries. Coastal development planning—including a Blue Economy—must integrate these communities, at the very least to minimize conflict between resource uses. Second, fisheries are more than the sum of their parts. Fisheries provide substantial benefits across the oceans [92] but are crucial at the community level, particularly when accounting for informal livelihoods [104,105], nutritional requirements [106], and Indigenous practices [95] that don't always appear in official statistics [107]. Third, there is a vast body of research and a growing set of strategies for improving fisheries management in a range of contexts that are directly applicable to other marine sectors [23,108]. While specific policies vary, most include effective measures against over exploitation and habitat degradation, equitable and inclusive access frameworks, and adaptive strategies to deal with environmental and economic changes; these strengthen ecosystem and social systems, benefitting any development plan.

Some of these strategies have recently been integrated into the FAO Sustainable Guidelines for Small-Scale Fisheries in the Context of Food

Security and Poverty Alleviation [25], which call for full inclusion and participation of seafood workers (throughout the value chain and emphatically including women) in policy development and management, with respect for cultures and traditions, equitable development, and transparency as key guidelines. Basic human rights are clearly central here, and indeed the UN Declaration on Human Rights can be linked to fisheries management at an operational level through an explicit focus on guaranteeing access to food, decent work, and the absence of forced labor or other forms of human insecurity [15]. This aligns well with a Blue Economy focused on social equity given that these objectives do not imply or require an expansion of fishing activity or production, but rather a re-evaluation of the many types of current and potential benefits and values of fisheries at local and global scales [7].

1.1.6. Mariculture

Aquaculture is one of the fastest growing forms of food fish production and, though contributions to protein supply are sometimes overstated [109], currently accounts for about 50% (80 million tonnes) of global yearly seafood production and US$232 billion in value [92]. Some 35% of aquaculture production (so, 17% of total food fish production) is from mariculture (including marine and brackish waters), a sub-sector whose recent growth is highly influenced by the increasing demand for seafood in developed countries, often with a focus on expensive species [92,110]. Most current production growth has been of shrimps in India and Vietnam, but there have been significant increases in production of salmons, mussels, and cobia in other regions of the world [110]. In addition to food, mariculture provides many inputs for industrial uses, and seaweed mariculture could support large-scale bioenergy production [111].

Under a Blue Economy, mariculture development will require a strategic approach that integrates operations within a wider bionetwork, e.g., through multitrophic farms, to provide benefits to local communities without compromising environmental sustainability. This is particularly important given observed impacts from pollution and diseases leading to economic losses for smaller mariculture farms, artisanal fisheries, and ecotourism in surrounding areas. An unresolved issue for future mariculture growth is the goal-specific planning and development of new farms, particularly as prioritizing mariculture expansion for local food consumption or livelihoods may require distinct characteristics than maximizing overall profits or increasing national competitiveness in seafood markets. Very large mariculture operations may be more economically efficient given their focus on high-value species, but they likely would not aim to maximize local employment, minimize impacts to other resource users and local residents, or produce food that is affordable for local communities. A Blue Economy approach would need to carefully consider the selection of species given objectives and environmental conditions, securing of cost-effective supplies of aquafeed, selection of areas to avoid social impacts, and technology and support facilities to produce at local affordable prices. There are potential synergies between mariculture and other sectors that could prove useful here, for example, by using existing infrastructure—including offshore wind farms—to support farming of bivalves and algae [112], and by combining mariculture and artisanal fisheries activities on a seasonal basis.

There are notable community-based mariculture initiatives involving production for local objectives with specific sustainability standards that span production chain including hatcheries, farms processors, feed mills and traders. Some examples include the Shrimp Seal of Quality (SSOQ), a certification program with guidelines for all stakeholders in the Bangladesh shrimp industry [113], the Malaysian Aquaculture Farm Certification Scheme (SPLAM) and the Code of Good Environmental Practices certification by SIGES-Salmon in Chile [114]. Despite their potential environmental benefits (e.g., limiting pollution, diseases, and introduction of invasive species), these programs mainly focus on acceptability of farm produce for export with limited consideration of local food security or livelihoods. Guidelines for social responsibility in aquaculture have more recently been incorporated in the Aquaculture Stewardship Council [114] which has certified a growing number of farms around the world, though governments ultimately have the responsibility to ensure that development is socially equitable.

Table 1
Examples of existing guidelines and key requirements for establishing examples of Blue Economy sectors in a socially equitable and environmentally sustainable manner.

Sector	Requirements	Equity and Sustainability Guidelines
Enabling Conditions (All Sectors)	• Recognition and inclusion in planning of all parties affected by a sector; • Improving well-being among local stakeholders; • Local capacity to benefit from high-skill employment; • Control of multi-level corruption; • Infrastructure and investor protection; • Adaptation strategies to environmental and economic changes.	UN SDG, UNDRIP, IPCC, UNUDHR, Green Economy Report, UNFCCC, UN GPA-Marine, Commonwealth Blue Charter
Bioprospecting	• Access to biodiversity; • Technological and analytical capacity.	CBD, Nagoya Protocol
Blue Carbon	• Abundant and healthy coastal plant systems; • Market demand for blue carbon investments.	REDD+, FCPF, Ramsar, LULUCF, Blue Bonds
Blue Energy	• Reliable wind and oceanographic patterns; • Shallow nearshore sites (current tech); • Users relatively close-by.	Green Economy, GP WIND, BOEM Guidelines
Ecotourism	• High biodiversity and/or species abundance; • Local infrastructure, safety and security; • Sustainable coastal development.	UNWTO, UNGA Ecotourism Resolution
Fisheries	• Effective measures against overfishing; • Equitable and inclusive access to fish stocks.	SSF Guidelines, EAF Mgmt, Fair Trade CFS, Seafish RFS
Mariculture	• Suitable areas for production; • Minimal impacts on surrounding ecosystems.	BAP, Ecosystem Approach to Aquaculture (EAA), ASC

1.2. Conclusions: transitioning to a Blue Economy

Global challenges including poverty, food security, climate change, and marginalization of various human groups require that we incorporate social issues in science and planning just as we have environmental ones. The SDGs cemented this need [3] and, for the world's oceans, the Blue Economy can be instrumental in facilitating this transformation. However, development is often shaped by objectives and discourses from the private sector or funding agencies with a focus on economic growth [5,6,115] and large NGOs focused on environmental conservation (e.g. Ref. [116]). It is therefore important that frameworks for a Blue Economy also explicitly address social inclusion and equitable outcomes [117], including protecting human rights, access to resources, and ensuring an equitable share of benefits and costs [118].

Most quantitative analyses of outcomes from development or particular sectors focus on measuring the distribution of financial benefits or costs among stakeholder groups, but not the prior levels of

recognition, inclusion, or participation, of stakeholders within the sector or development process [20]. In that sense, including social equity as an active objective of a Blue Economy helps make planning trade-offs clear and prompts the design of development, implementation, and monitoring plans to allow for evaluating equity considerations throughout the process. As noted in the sector reviews, there are various high-level agreements, documents, and codes of practice that are already recognized and in use, offering guidance on how such an integration can be started in specific contexts. We do not aim to provide a formal review of this literature, but we summarize the key guidelines mentioned above in Table 1.

There are methods to more adequately monitor social aspects of development, including participation in industries, perceptions of impacts, and distributions of costs and benefits between locations or human groups [19]. More importantly, it is important that underlying governance contexts support improved inclusion, recognition, and equitable outcomes in marine planning. This can include an emphasis on a wide consideration of stakeholders, strengthened legal frameworks to formally institute co-management arrangements that devolve some power to stakeholders, conceptualization (within policy) of marine sectors as social-ecological systems, the integration of operational realities in planning, and a strong focus on transparency and information sharing [19,119]. Local communities should also have the opportunity to access benefits from new industries, not only existing or "traditional" ones. In some cases, existing industries may need to be downscaled if they are unsustainable as currently carried out, but the strategies by which this is done can concurrently achieve a more equitable distribution of subsequent benefits, sometimes with a net gain for local communities [19].

Environmental guidelines have now been largely incorporated—at the very least nominally—in virtually any development plan, but social and human rights considerations have lagged far behind [15]. There is a clear need and opportunity for the 'Blue Economy' to represent a truly novel development framework and avoid a rebranding of business-as-usual marine operations that continue to harm marine ecosystems and human communities around the world. A Blue Economy that makes social equity the cornerstone of a new approach to development may prove instrumental to achieving sustainability and well-being in all its dimensions.

Acknowledgements

AMCM, MMB, WWLC, MO, GGS, WS, and YO acknowledge support from the Nippon Foundation Nereus Program, a research partnership between the Nippon Foundation and academic partners around the world.

References

[1] J.J. Silver, N.J. Gray, L.M. Campbell, L.W. Fairbanks, R.L. Gruby, Blue economy and competing discourses in international oceans governance, J. Environ. Dev. 24 (2015) 135–160, https://doi.org/10.1177/1070496515580797.
[2] UNEP, Driving a Green Economy through Public Finance and Fiscal Policy Reform, United Nations Environment Programme, 2010.
[3] UN, Transforming Our World: the 2030 Agenda for Sustainable Development, United Nations, New York, 2015.
[4] K.S. Park, J.T. Kildow, Rebuilding the classification system of the ocean economy, J. Ocean Coast. Econ. 2014 (2015), https://doi.org/10.15351/2373-8456.1001.
[5] Z.W. Brent, M. Barbesgaard, C. Pedersen, The Blue Fix: Unmasking the Politics behind the Promise of Blue Growth, Transnational Institute, 2018.
[6] M. Voyer, G. Quirk, A. McIlgorm, K. Azmi, Shades of blue: what do competing interpretations of the Blue Economy mean for oceans governance? J. Environ. Policy Plan. (2018) 1–22, https://doi.org/10.1080/1523908X.2018.1473153.
[7] W.J. Boonstra, M. Valman, E. Björkvik, A sea of many colours – how relevant is Blue Growth for capture fisheries in the Global North, and vice versa? Mar. Policy 87 (2018) 340–349, https://doi.org/10.1016/j.marpol.2017.09.007.
[8] M. Voyer, J. van Leeuwen, Social License to Operate and the Blue Economy. Report to World Ocean Council, Australian Centre for Ocean Resources and Security, Wollongong, Australia, 2018.
[9] WFFP, Fisher Peoples Reject the "Our Ocean Conference" and Organize the Ocean's People Conference, 2018.
[10] N.J. Bennett, H. Govan, T. Satterfield, Ocean grabbing, Mar. Policy 57 (2015) 61–68, https://doi.org/10.1016/j.marpol.2015.03.026.
[11] OECD, The Ocean Economy in 2030, OECD Publishing, 2016, https://doi.org/10.1787/9789264251724-en.
[12] L.C.L. Teh, U.R. Sumaila, Contribution of marine fisheries to worldwide employment: global marine fisheries employment, Fish Fish. 14 (2013) 77–88, https://doi.org/10.1111/j.1467-2979.2011.00450.x.
[13] ECORYS, Deltares, Oceanic Développement. Blue Growth. Scenarios and Drivers for Sustainable Growth from the Oceans. Seas and Coasts, 2012.
[14] Economist Intelligence Unit, The Blue Economy. Growth, Opportunity and a Sustainable Ocean Economy, The Economist, 2015.
[15] L.C.L. Teh, R. Caddell, E.H. Allison, E.M. Finkbeiner, J.N. Kittinger, K. Nakamura, et al., The role of human rights in implementing socially responsible seafood, PLoS One 14 (2019), e0210241, https://doi.org/10.1371/journal.pone.0210241.
[16] IPCC, Climate Change 2014: Synthesis Report. Contribution of Working Groups I, II and III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change, IPCC, Geneva, 2014.
[17] World Bank, Poverty and Shared Prosperity 2016: Taking on Inequality, The World Bank, 2016, https://doi.org/10.1596/978-1-4648-0958-3.
[18] U. Pascual, J. Phelps, E. Garmendia, K. Brown, E. Corbera, A. Martin, et al., Social equity matters in payments for ecosystem Services, Bioscience 64 (2014) 1027–1036, https://doi.org/10.1093/biosci/biu146.
[19] N.J. Bennett, J. Blythe, A.M. Cisneros-Montemayor, G.G. Singh, U.R. Sumaila, Just transformations to sustainability, Sustainability 11 (2019) 3881, https://doi.org/10.3390/su11143881.
[20] B.S. Halpern, C.J. Klein, C.J. Brown, M. Beger, H.S. Grantham, S. Mangubhai, et al., Achieving the triple bottom line in the face of inherent trade-offs among social equity, economic return, and conservation, Proc. Natl. Acad. Sci. 110 (2013) 6229–6234, https://doi.org/10.1073/pnas.1217689110.
[21] P.G. Patil, J. Virdin, C.S. Colgan, M.G. Hussain, P. Failler, T. Vegh, Toward a Blue Economy: A Pathway for Bangladesh's Sustainable Growth, World Bank, Washington, DC, 2018.
[22] World Bank, The Sunken Billions Revisited: Progress and Challenges in Global Marine Fisheries, The World Bank, 2017, https://doi.org/10.1596/978-1-4648-0919-4.
[23] U.R. Sumaila, W. Cheung, A. Dyck, K. Gueye, L. Huang, V. Lam, et al., Benefits of rebuilding global marine fisheries outweigh costs, PLoS One 7 (2012), e40542, https://doi.org/10.1371/journal.pone.0040542.
[24] N.J. Bennett, Navigating a just and inclusive path towards sustainable oceans, Mar. Policy 97 (2018) 139–146, https://doi.org/10.1016/j.marpol.2018.06.001.
[25] FAO, Voluntary Guidelines for Securing Sustainable Small-Scale Fisheries in the Context of Food Security and Poverty Eradication, Food and Agriculture Organization of the United Nations, Rome, 2015.
[26] H. Tallis, P.S. Levin, M. Ruckelshaus, S.E. Lester, K.L. McLeod, D.L. Fluharty, et al., The many faces of ecosystem-based management: making the process work today in real places, Mar. Policy 34 (2010) 340–348, https://doi.org/10.1016/j.marpol.2009.08.003.
[27] B. Hunt, A.C.J. Vincent, Scale and sustainability of marine bioprospecting for pharmaceuticals, AMBIO A J. Hum. Environ. 35 (2006) 57–64, https://doi.org/10.1579/0044-7447(2006)35[57:SASOMB]2.0.CO;2.
[28] A.M. Cisneros-Montemayor, U.R. Sumaila, A global estimate of benefits from ecosystem-based marine recreation: potential impacts and implications for management, J. Bioecon. 12 (2010) 245–268.
[29] W.W.L. Cheung, V.W.Y. Lam, J.L. Sarmiento, K. Kearney, R. Watson, D. Zeller, et al., Large-scale redistribution of maximum fisheries catch potential in the global ocean under climate change, Glob. Chang. Biol. 16 (2010) 24–35, https://doi.org/10.1111/j.1365-2486.2009.01995.x.
[30] M.A. Oyinlola, G. Reygondeau, C.C. Wabnitz, M. Troell, W.W. Cheung, Global estimation of areas with suitable environmental conditions for mariculture species, PLoS One 13 (2018), e0191086.
[31] S. Thomas, Blue carbon: knowledge gaps, critical issues, and novel approaches, Ecol. Econ. 107 (2014) 22–38, https://doi.org/10.1016/j.ecolecon.2014.07.028.
[32] C wei Zheng, J. Pan, Assessment of the global ocean wind energy resource, Renew. Sustain. Energy Rev. 33 (2014) 382–391, https://doi.org/10.1016/j.rser.2014.01.065.
[33] World Bank, UN-DESA, The Potential of the Blue Economy. Increasing Long-Term Benefits of the Sustainable Use of Marine Resources for Small Island Developing States and Coastal Least Developed Countries, World Bank, Washington, D.C., 2017.
[34] D. Griggs, M. Stafford-Smith, O. Gaffney, J. Rockström, M.C. Öhman, P. Shyamsundar, et al., Sustainable development goals for people and planet: Policy, Nature 495 (2013) 305–307, https://doi.org/10.1038/495305a.
[35] D. Farrier, L. Tucker, Access to marine bioresources: hitching the conservation cart to the bioprospecting horse, Ocean Dev. Int. Law 32 (2001) 213–239, https://doi.org/10.1080/009083201750397583.
[36] R.C. Pereira, L.V. Costa-Lotufo, Bioprospecting for bioactives from seaweeds: potential, obstacles and alternatives, Rev. Bras. Farmacogn. 22 (2012) 894–905.
[37] G.-P. Hu, J. Yuan, L. Sun, Z.-G. She, J.-H. Wu, X.-J. Lan, et al., Statistical research on marine natural products based on data obtained between 1985 and 2008, Mar. Drugs 9 (2011) 514–525, https://doi.org/10.3390/md9040514.
[38] R. Blasiak, J.-B. Jouffray, C.C. Wabnitz, E. Sundström, H. Österblom, Corporate control and global governance of marine genetic resources, Sci. Adv. 4 (2018) eaar5237.
[39] A.R. Haas, D.N. Edwards, U.R. Sumaila, Corporate concentration and processor control: insights from the salmon and herring fisheries in British Columbia, Mar. Policy 68 (2016) 83–90, https://doi.org/10.1016/j.marpol.2016.02.019.

[40] U.N. Convention, On Biological Diversity, Rio de Janeiro, 1992.
[41] Y. Demunshi, A. Chugh, Role of traditional knowledge in marine bioprospecting, Biodivers. Conserv. 19 (2010) 3015–3033, https://doi.org/10.1007/s10531-010-9879-9.
[42] R. Blasiak, J. Pittman, N. Yagi, H. Sugino, Negotiating the use of biodiversity in marine areas beyond national jurisdiction, Front. Mar. Sci. 3 (2016), https://doi.org/10.3389/fmars.2016.00224.
[43] C. Nellemann, E. Corcoran, Blue Carbon: the Role of Healthy Oceans in Binding Carbon : a Rapid Response Assessment, UNEP/Earthprint, 2009.
[44] C.M. Duarte, H. Kennedy, N. Marbà, I. Hendriks, Assessing the capacity of seagrass meadows for carbon burial: current limitations and future strategies, Ocean Coast Manag. 83 (2013) 32–38, https://doi.org/10.1016/j.ocecoaman.2011.09.001.
[45] C.M. Duarte, N. Marbà, E. Gacia, J.W. Fourqurean, J. Beggins, C. Barrón, et al., Seagrass community metabolism: assessing the carbon sink capacity of seagrass meadows, Glob. Biogeochem. Cycles 24 (2010) GB4032, https://doi.org/10.1029/2010GB003793.
[46] G.L. Chmura, S.C. Anisfeld, D.R. Cahoon, J.C. Lynch, Global carbon sequestration in tidal, saline wetland soils, Glob. Biogeochem. Cycles 17 (2003), https://doi.org/10.1029/2002GB001917.
[47] S. Bouillon, A.V. Borges, E. Castañeda-Moya, K. Diele, T. Dittmar, N.C. Duke, et al., Mangrove production and carbon sinks: a revision of global budget estimates, Glob. Biogeochem. Cycles 22 (2008), https://doi.org/10.1029/2007GB003052.
[48] H. Kennedy, J. Beggins, C.M. Duarte, J.W. Fourqurean, M. Holmer, N. Marbà, et al., Seagrass sediments as a global carbon sink: isotopic constraints, Glob. Biogeochem. Cycles 24 (2010), https://doi.org/10.1029/2010GB003848.
[49] E. Mcleod, G.L. Chmura, S. Bouillon, R. Salm, M. Björk, C.M. Duarte, et al., A blueprint for blue carbon: toward an improved understanding of the role of vegetated coastal habitats in sequestering CO2, Front. Ecol. Environ. 9 (2011) 552–560, https://doi.org/10.1890/110004.
[50] M. Diesing, S. Kröger, R. Parker, C. Jenkins, C. Mason, K. Weston, Predicting the standing stock of organic carbon in surface sediments of the North–West European continental shelf, Biogeochemistry 135 (2017) 183–200, https://doi.org/10.1007/s10533-017-0310-4.
[51] T. Luisetti, R.K. Turner, J.E. Andrews, T.D. Jickells, S. Kröger, M. Diesing, et al., Quantifying and valuing carbon flows and stores in coastal and shelf ecosystems in the UK, Ecosyst. Serv. 35 (2019) 67–76, https://doi.org/10.1016/j.ecoser.2018.10.013.
[52] J.E. Fargione, S. Bassett, T. Boucher, S.D. Bridgham, R.T. Conant, S.C. Cook-Patton, et al., Natural climate solutions for the United States, Sci. Adv. 4 (2018), eaat1869, https://doi.org/10.1126/sciadv.aat1869.
[53] L. Pendleton, D.C. Donato, B.C. Murray, S. Crooks, W.A. Jenkins, S. Sifleet, et al., Estimating global "blue carbon" emissions from conversion and degradation of vegetated coastal ecosystems, PLoS One 7 (2012), e43542, https://doi.org/10.1371/journal.pone.0043542.
[54] A.P. Hejnowicz, H. Kennedy, M.A. Rudd, M.R. Huxham, Harnessing the climate mitigation, conservation and poverty alleviation potential of seagrasses: prospects for developing blue carbon initiatives and payment for ecosystem service programmes, Front. Mar. Sci. 2 (2015), https://doi.org/10.3389/fmars.2015.00032.
[55] D. Herr, E. Pidgeon, Climate Change Mitigation Activities in Coastal Ecosystems; Recommendations for the UNFCCC Processes, Internationla Blue Carbon Policy Working Group, 2012.
[56] D. Herr, M. von Unger, D. Laffoley, A. McGivern, Pathways for implementation of blue carbon initiatives, Aquat. Conserv. Mar. Freshw. Ecosyst. 27 (2017) 116–129, https://doi.org/10.1002/aqc.2793.
[57] R. Ullman, V. Bilbao-Bastida, G. Grimsditch, Including Blue Carbon in climate market mechanisms, Ocean Coast Manag. 83 (2013) 15–18, https://doi.org/10.1016/j.ocecoaman.2012.02.009.
[58] D. Herr, E. Landis, Coastal Blue Carbon Ecosystems, Opportunities for Nationally Determined Contributions, 2016.
[59] J. Howard, E. McLeod, S. Thomas, E. Eastwood, M. Fox, L. Wenzel, et al., The potential to integrate blue carbon into MPA design and management, Aquat. Conserv. Mar. Freshw. Ecosyst. 27 (2017) 100–115, https://doi.org/10.1002/aqc.2809.
[60] J. Howard, A. Sutton-Grier, D. Herr, J. Kleypas, E. Landis, E. Mcleod, et al., Clarifying the role of coastal and marine systems in climate mitigation, Front. Ecol. Environ. 15 (2017) 42–50, https://doi.org/10.1002/fee.1451.
[61] H. Schroeder, C. McDermott, Beyond carbon: enabling justice and equity in REDD + across levels of governance, Ecol. Soc. 19 (2014), https://doi.org/10.5751/ES-06537-190131.
[62] M. Menton, C. Ferguson, R. Leimu-Brown, S. Leonard, M. Brockhaus, A. E. Duchelle, et al., Further Guidance for REDD+ Safeguard Information Systems?: an Analysis of Positions in the UNFCCC Negotiations, CIFOR, 2014.
[63] L. Wylie, A.E. Sutton-Grier, A. Moore, Keys to successful blue carbon projects: lessons learned from global case studies, Mar. Policy 65 (2016) 76–84, https://doi.org/10.1016/j.marpol.2015.12.020.
[64] J. Howard, S. Hoyt, K. Isensee, M. Telszewski, E. Pidgeon (Eds.), Coastal Blue Carbon: Methods for Assessing Carbon Stocks and Emissions Factors in Mangroves, tidal salt marshes, and seagrasses, 2014.
[65] K. Benessaiah, Carbon and livelihoods in Post-Kyoto: assessing voluntary carbon markets, Ecol. Econ. 77 (2012) 1–6, https://doi.org/10.1016/j.ecolecon.2012.02.022.
[66] M. Peters-Stanley, D. Yin, Maneuvering the Mosaic; State of the Voluntary Carbon Markets, Forest Trends's Ecosystem Marketplace A report by Forest Trend's Ecosystem Marketplace & Bloomberg New Energy Finance, 2013, 2013.
[67] A. Kollmuss, H. Zink, C. Polycarp, Making Sense of the Voluntary Carbon Market: a Comparison of Carbon Offset Standards, WWF Germany, 2008.
[68] M.E. Röhr, M. Holmer, J.K. Baum, M. Björk, K. Boyer, D. Chin, et al., Blue carbon storage capacity of temperate eelgrass (Zostera marina) meadows, Glob. Biogeochem. Cycles 32 (2018) 1457–1475, https://doi.org/10.1029/2018GB005941.
[69] C. Prentice, R.D. Smith, M. Hessing-Lewis, A.K. Salomon, Reduced water motion enhances carbon stocks in temperate eelgrass meadows, Limnol. Oceanogr. (2019), https://doi.org/10.1002/lno.11191 submitted for publication.
[70] K.L. Poppe, J.M. Rybczyk, Carbon sequestration in a pacific northwest eelgrass (Zostera marina) meadow, Northwest Sci. 92 (2018) 80–91, https://doi.org/10.3955/046.092.0202.
[71] CIESIN, Percentage of Total Population Living in Coastal Areas, 2006.
[72] U. Chakravorty, K. Emerick, M.-L. Ravago, Lighting up the last mile: the benefits and costs of extending electricity to the rural poor, SSRN Electron. J. (2016), https://doi.org/10.2139/ssrn.2851907.
[73] A. Bermudez-Contreras, M. Thomson, D.G. Infield, Renewable energy powered desalination in Baja California Sur, Mexico, Desalination 220 (2008) 431–440, https://doi.org/10.1016/j.desal.2007.01.046.
[74] E. Denny, The economics of tidal energy, Energy Policy 37 (2009) 1914–1924, https://doi.org/10.1016/j.enpol.2009.01.009.
[75] S.-P. Breton, G. Moe, Status, plans and technologies for offshore wind turbines in Europe and North America, Renew. Energy 34 (2009) 646–654, https://doi.org/10.1016/j.renene.2008.05.040.
[76] A. O'Keeffe, C. Haggett, An investigation into the potential barriers facing the development of offshore wind energy in Scotland: case study – firth of Forth offshore wind farm, Renew. Sustain. Energy Rev. 16 (2012) 3711–3721, https://doi.org/10.1016/j.rser.2012.03.018.
[77] S.C. Klain, T. Satterfield, S. MacDonald, N. Battista, K.M.A. Chan, Will communities "open-up" to offshore wind? Lessons learned from New England islands in the United States, Energy Res. Soc. Sci. 34 (2017) 13–26, https://doi.org/10.1016/j.erss.2017.05.009.
[78] R. Ballantyne, J. Packer, K. Hughes, Tourists' support for conservation messages and sustainable management practices in wildlife tourism experiences, Tour. Manag. 30 (2009) 658–664, https://doi.org/10.1016/j.tourman.2008.11.003.
[79] A.J. Gallagher, N. Hammerschlag, Global shark currency: the distribution, frequency, and economic value of shark ecotourism, Curr. Issues Tourism 14 (2011) 797–812, https://doi.org/10.1080/13683500.2011.585227.
[80] S. O'Connor, R. Campbell, H. Cortez, T. Knowles, Whale Watching Worldwide: Tourism Numbers, Expenditures and Expanding Economic Benefits, a Special Report from the International Fund for Animal Welfare, International Fund for Animal Welfare, Yarmouth, 2009. Yarmouth, USA, prepared by Economists at Large.
[81] M.M. Khan, Tourism development and dependency theory: mass tourism vs. ecotourism, Ann. Tourism Res. 24 (1997) 988–991.
[82] M. Honey, D. Krantz, Global trends in coastal tourism, Report prepared for World Wildlife Fund, Washington DC (2007) 140.
[83] A.M. Cisneros-Montemayor, U.R. Sumaila, K. Kaschner, D. Pauly, The global potential for whale watching, Mar. Policy (2010) 1273–1278.
[84] C.C.C. Wabnitz, A.M. Cisneros-Montemayor, Q. Hanich, Y. Ota, Ecotourism, climate change and reef fish consumption in Palau: benefits, trade-offs and adaptation strategies, Mar. Policy (2017), https://doi.org/10.1016/j.marpol.2017.07.022.
[85] H.M. Donohoe, R.D. Needham, Ecotourism: the evolving contemporary definition, J. Ecotourism 5 (2006) 192–210, https://doi.org/10.2167/joe152.0.
[86] V.J. Harriott, Marine Tourism Impacts and Their Management on the Great Barrier Reef, CRC Reef Research Centre Townesville, QLD, Australia, 2002.
[87] N. Hammerschlag, A.J. Gallagher, J. Wester, J. Luo, J.S. Ault, Don't bite the hand that feeds: assessing ecological impacts of provisioning ecotourism on an apex marine predator: ecological impacts of shark ecotourism, Funct. Ecol. 26 (2012) 567–576, https://doi.org/10.1111/j.1365-2435.2012.01973.x.
[88] G. de F. Filla, A. Monteiro-Filho EL de, Monitoring tourism schooners observing estuarine dolphins (*Sotalia guianensis*) in the Estuarine Complex of Cananéia, south-east Brazil, Aquat. Conserv. Mar. Freshw. Ecosyst. 19 (2009) 772–778, https://doi.org/10.1002/aqc.1034.
[89] M.B. Orams, Tourists getting close to whales, is it what whale-watching is all about? Tour. Manag. 21 (2000) 561–569.
[90] P.F.J. Eagles, S.F. McCool, Sustainable Tourism in Protected Area: Guidelines for Planning and Management, World Conservation Union, 2003.
[91] D.B. Weaver, L.J. Lawton, Twenty years on: the state of contemporary ecotourism research, Tour. Manag. 28 (2007) 1168–1179, https://doi.org/10.1016/j.tourman.2007.03.004.
[92] FAO, The State of World Fisheries and Aquaculture 2018, Meeting the Sustainable Development Goals, Rome, 2018.
[93] C. Bellmann, A. Tipping, U.R. Sumaila, Global trade in fish and fishery products: an overview, Mar. Policy 69 (2016) 181–188, https://doi.org/10.1016/j.marpol.2015.12.019.
[94] S. Schumann, S. Macinko, Subsistence in coastal fisheries policy: what's in a word? Mar. Policy 31 (2007) 706–718, https://doi.org/10.1016/j.marpol.2006.12.010.
[95] A.M. Cisneros-Montemayor, D. Pauly, L.V. Weatherdon, Y. Ota, A global estimate of seafood consumption by coastal indigenous peoples, PLoS One 11 (2016), e0166681, https://doi.org/10.1371/journal.pone.0166681.
[96] European Commission, Blue Growth Opportunities for Marine and Maritime Sustainable Growth, European Commission, Brussels, 2012.

[97] G. Ortuño Crespo, D.C. Dunn, A review of the impacts of fisheries on open-ocean ecosystems, ICES J. Mar. Sci. (2017), https://doi.org/10.1093/icesjms/fsx084.
[98] J.B.C. Jackson, Historical overfishing and the recent collapse of coastal ecosystems, Science 293 (2001) 629–637, https://doi.org/10.1126/science.1059199.
[99] A.M. Cisneros-Montemayor, E. Sanjurjo, G.R. Munro, V. Hernández-Trejo, U. R. Sumaila, Strategies and rationale for fishery subsidy reform, Mar. Policy (2016) 229–236, https://doi.org/10.1016/j.marpol.2015.10.001.
[100] U.R. Sumaila, Trade Policy Options for Sustainable Oceans and Fisheries, International Centre for Trade and Sustainable Development (ICTSD) and World Economic Forum, Geneva, 2016.
[101] J.S. Brashares, B. Abrahms, K.J. Fiorella, C.D. Golden, R.A. Marsh, D.J. McCauley, et al., Wildlife decline and social conflict, Science 345 (2014) 376–378.
[102] E. Bennett, A. Neiland, E. Anang, P. Bannerman, A.A. Rahman, S. Huq, et al., Towards a better understanding of conflict management in tropical fisheries: evidence from Ghana, Bangladesh and the Caribbean, Mar. Policy 25 (2001) 365–376.
[103] U.R. Sumaila, V. Lam, F. Le Manach, W. Swartz, D. Pauly, Global fisheries subsidies: an updated estimate, Mar. Policy 69 (2016) 189–193, https://doi.org/10.1016/j.marpol.2015.12.026.
[104] N. Guillemot, M. Léopold, M. Cuif, P. Chabanet, Characterization and management of informal fisheries confronted with socio-economic changes in New Caledonia (South Pacific), Fish. Res. 98 (2009) 51–61, https://doi.org/10.1016/j.fishres.2009.03.013.
[105] A.M. Cisneros-Montemayor, S. Harper, T.C. Tai, The market and shadow value of informal fish catch: a framework and application to Panama: Andrés Cisneros-Montemayor, Sarah S. Harper and Travis Tai/Natural Resources Forum, Nat. Resour. Forum (2018), https://doi.org/10.1111/1477-8947.12143.
[106] C. Golden, E.H. Allison, W.W. Cheung, M.M. Dey, B.S. Halpern, D.J. McCauley, et al., Fall in fish catch threatens human health, Nature 317–20 (2016).
[107] D. Pauly, D. Zeller (Eds.), Global Atlas of Marine Fisheries. A Critical Appraisal of Catches and Ecosystem Impacts, Island Press, Washington, D.C., 2016.
[108] C. Costello, D. Ovando, R. Hilborn, S.D. Gaines, O. Deschenes, S.E. Lester, Status and solutions for the world's unassessed fisheries, Science 338 (2012) 517–520, https://doi.org/10.1126/science.1223389.
[109] P. Edwards, W. Zhang, B. Belton, D.C. Little, Misunderstandings, myths and mantras in aquaculture: its contribution to world food supplies has been systematically over reported, Mar. Policy 106 (2019) 103547, https://doi.org/10.1016/j.marpol.2019.103547.
[110] B. Campbell, D. Pauly, Mariculture: a global analysis of production trends since 1950, Mar. Policy 39 (2013) 94–100, https://doi.org/10.1016/j.marpol.2012.10.009.
[111] S. Kraan, Mass-cultivation of carbohydrate rich macroalgae, a possible solution for sustainable biofuel production, Mitig. Adapt. Strategies Glob. Change 18 (2013) 27–46, https://doi.org/10.1007/s11027-010-9275-5.
[112] T. Michler-Cieluch, G. Krause, B.H. Buck, Reflections on integrating operation and maintenance activities of offshore wind farms and mariculture, Ocean Coast Manag. 52 (2009) 57–68, https://doi.org/10.1016/j.ocecoaman.2008.09.008.
[113] M.S. Islam, From sea to shrimp processing factories in Bangladesh: gender and employment at the bottom of a global commodity chain, J. South Asian Dev. 3 (2008) 211–236, https://doi.org/10.1177/097317410800300202.
[114] B.E. Cid Aguayo, J.M. Barriga Parra, Governances and invisibilities: interests and rationalities in the socio-environmental regulation of salmon farming in Chile, Ambiente Sociedade 20 (2017) 105–120, https://doi.org/10.1590/1809-4422asoc261r2v2022017.
[115] F. Guerra, Mapping offshore renewable energy governance, Mar. Policy 89 (2018) 21–33, https://doi.org/10.1016/j.marpol.2017.12.006.
[116] O. Hoegh-Guldberg, D. Beal, T. Chaudhry, H. Elhaj, A. Abdullat, P. Etessy, et al., Reviving the Ocean Economy: the Case for Action - 2015, WWF International, Gland, Switzerland, 2015.
[117] M.R. Keen, A.-M. Schwarz, L. Wini-Simeon, Towards defining the Blue Economy: practical lessons from pacific ocean governance, Mar. Policy (2017), https://doi.org/10.1016/j.marpol.2017.03.002.
[118] J.N. Kittinger, L.C. Teh, E.H. Allison, N.J. Bennett, L.B. Crowder, E.M. Finkbeiner, et al., Committing to socially responsible seafood, Science 356 (2017) 912–913.
[119] R.L. Stephenson, A.J. Benson, K. Brooks, A. Charles, P. Degnbol, C.M. Dichmont, et al., Practical steps toward integrating economic, social and institutional elements in fisheries policy and management, ICES J. Mar. Sci. 74 (2017) 1981–1989, https://doi.org/10.1093/icesjms/fsx057.